

Main Tools of a Data Scientist

- **Programming and Collaboration Tools:** LaTeX/Markdown, Git/GitHub, Linux command line, SSH
- **Programming Languages for Data Collection and Statistical Modeling:** R, Julia, Python
- **Big Data Management:** SQL, RDDs (Resilient Distributed Datasets) including Spark and Hadoop
- **High-Performance Computing Resources:** SSH (Secure Shell), SFTP (Secure File Transfer Protocol), SLURM (Simple Linux Utility for Resource Management)
- **Data Collection Methods:** APIs, Web scraping
- **Data Cleaning and Visualization:** Best practices for preparing and visualizing data
- **Computational Methods:** Optimization, root finding, Monte Carlo simulations, parallel computing
- **Machine Learning:** Supervised and unsupervised learning, model training and validation
- **Economic Modeling:** Computing structural models for policy analysis